Contents

Chapter 5

The Things Topology Are

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5.3 Problem 5.3

For each of the following, provide a proof or a counterexample.

(a) If A and B are compact, must $A \cup B$ be compact?

(b) If A and B are compact, must $A \cap B$ be compact?

5.3.1 Solution (a)

Proof. By the Heine-Borel Theorem, if A and B are compact, the two of them are closed and bounded. We can name bounds for the two of these and recall the definition of being closed (having an open complement).

$$\forall a \in A, L_A \leq a \leq U_A \tag{5.1}$$
$$\forall b \in B, L_B \leq b \leq U_B \tag{5.2}$$
$$\forall x \in A^c, \exists \delta > 0 : (x - \delta, x + \delta) \subseteq A^c \tag{5.3}$$
$$\forall x \in B^c, \exists \delta > 0 : (x - \delta, x + \delta) \subseteq B^c \tag{5.4}$$

Suppose we define the union of A and B and its complement by DeMorgan's Law.

$$A \cup B = \{c | c \in A \vee c \in B\} \tag{5.5}$$
$$(A \cup B)^c = A^c \cap B^c \tag{5.6}$$

Since both A^c and B^c are open and the intersection of a finite number of open sets is also open, $A^c \cap B^c$ is open. By extension, $(A \cup B)^c$ is open and $A \cup B$ is closed.

Suppose that $U = \max(U_A, U_B)$ and $L = \min(L_A, L_B)$.

$$\forall a \in A, U \geq U_A \geq a \tag{5.7}$$
$$\forall b \in B, U \geq U_B \geq b \tag{5.8}$$
$$\forall a \in A, L \leq L_A \leq a \tag{5.9}$$
$$\forall b \in B, L \leq L_B \leq b \tag{5.10}$$

From this, we conclude that U is an upper bound of $A \cup B$ and L is a lower bound of $A \cup B$. This means that $A \cup B$ is bounded. Since $A \cup B$ is both closed and bounded, then by the Heine-Borel Theorem, it is compact. OEΔ

5.3.2 Solution (b)

Proof. By the Heine-Borel Theorem, if A and B are compact, the two of them are closed and bounded. This means that A^c and B^c are open. This means that $A^c \cup B^c$, which can be turned into $(A \cap B)^c$ by DeMorgan's Laws, is open. $A \cap B$ is closed.

We can write the definition of $A \cap B$ and name bounds for the two of these.

$$A \cap B = \{c | c \in A \text{ and } c \in B\} \tag{5.11}$$

$$\forall a \in A, L_A \leq a \leq U_A \tag{5.12}$$

$$\forall b \in B, L_B \leq b \leq U_B \tag{5.13}$$

$$\forall c \in A \cap B, L_A \leq c \leq U_A \text{ and } L_B \leq c \leq U_B \tag{5.14}$$

Since there is a minimum and maximum bound of c, we can say that $A \cap B$ is bounded.

Since a bounded and closed set is compact, $A \cap B$ is compact. OEΔ

5.4 Problem 5.5

(a) Give an example of countably many disjoint open intervals.

(b) Prove that there does not extst a collection of countably many disjoint open intervals.

5.5 Problem 5.7

5.6 Problem 5.21

For sets A and B, define $A \cdot B = a \cdot b : a \in A, b \in B$.

(a) If A and B are open, must $A \cdot B$ be open?

(b) If A and B are compact, must $A \cdot B$ be compact?

(c) If A and B are closed, must $A \cdot B$ be closed?

5.6.1 Solution (a)

Yes.

Proof. Recall what it means that A and B are open. For A, it means that for all $a \in A$, there exists some $\delta_a > 0$ such that $(a - \delta_a, a + \delta_a) \subseteq A$. For B, it means that for all $b \in B$, there exists some $\delta_b > 0$ such that $(b - \delta_b, b + \delta_b) \subseteq B$.

Suppose some term $a \in A$ and some term $b \in B$. We wind up with multiple cases, based on whether a or b or both is positive or negative, or if either is zero.

Case 1: Both positive. If both are positive, that means that $\frac{\delta_b}{a} > 0$.

$$\beta \in \left(b - \frac{\delta_b}{a}, b + \frac{\delta_b}{a}\right) \subseteq B \tag{5.15}$$

This can be rewritten, for all β.

$$b - \frac{\delta_b}{a} < \beta < b + \frac{\delta_b}{a} \tag{5.16}$$

Multiply all sides by a. Since $a \in A$ and $\beta \in B$, this means that $a\beta \in A \cdot B$.

$$ab - \delta_b < a\beta < ab + \delta_b \tag{5.17}$$

This means that $\forall a\beta \in (ab - \delta_b, ab + \delta_b)$, $a\beta \in A \cdot B$. It also means that $(ab - \delta_b, ab + \delta_b) \subseteq A \cdot B$. This means that $A \cdot B$ is open.

Case 2: Either or both negative. Suppose that just a is negative. If only b is negative, this can be changed appropriately. This means that $\frac{\delta_b}{a} < 0$, and that $-\frac{\delta_b}{a} > 0$.

$$\beta \in \left(b + \frac{\delta_b}{a}, b - \frac{\delta_b}{a}\right) \subseteq B \tag{5.18}$$

This can be rewritten, for all β.

$$b + \frac{\delta_b}{a} < \beta < b - \frac{\delta_b}{a} \tag{5.19}$$

Multiply all sides by a. This will reverse the signs. Since $a \in A$ and $\beta \in B$, this means that $a\beta \in A \cdot B$.

$$ab + \delta_b > a\beta > ab - \delta_b \tag{5.20}$$

This means that $\forall a\beta \in (ab - \delta_b, ab + \delta_b)$, $a\beta \in A \cdot B$. It also means that $(ab - \delta_b, ab + \delta_b) \subseteq A \cdot B$.

Case 3: Either are zero. Suppose that $a = 0$ and $b \neq 0$. We can rewrite our open interval.

$$\left(a - \frac{\delta_a}{b}, a + \frac{\delta_a}{b}\right) = \left(-\frac{\delta_a}{b}, \frac{\delta_a}{b}\right) \subseteq A \tag{5.21}$$

Multiply it by $b \in B$.

$$(-\delta_a, \delta_a) \subseteq A \cdot B \tag{5.22}$$

Case 4: Both are zero. Suppose that $a = 0$ and $b = 0$. Recall that $\delta_b > 0$, which also means $\frac{\delta_b}{2} > 0$. We can rewrite our open interval.

$$\left(a - \frac{\delta_a}{\delta_b}, a + \frac{\delta_a}{\delta_b}\right) = \left(-\frac{\delta_a}{\delta_b}, \frac{\delta_a}{\delta_b}\right) \subseteq A \tag{5.23}$$

Multiply it by $b + \frac{\delta_b}{2} = \frac{\delta_b}{2} \in B$.

$$\left(-\frac{\delta_a}{2}, \frac{\delta_a}{2}\right) \subseteq A \cdot B \tag{5.24}$$

This covers all the cases, which means that $A \cdot B$ is open. OEΔ

5.6.2 Solution (b)

Because A and B are compact, both are closed and bounded. This means that all $a \in A$ and $b \in B$ are within intervals.

$$L_A < a < U_A \tag{5.25}$$
$$L_B < b < U_B \tag{5.26}$$

We can multiply the terms together, and that establish a set of configurations depending on which terms are negative.

$$S = \{L_A \cdot L_B, U_A \cdot L_B, L_A \cdot U_B, U_A \cdot U_B\} \tag{5.27}$$
$$\min\{S\} < ab < \max\{S\} \tag{5.28}$$

This means $A \cdot B$ is bounded.

Suppose now that there is a sequence in $A \cdot B$ called (x_n). For all elements in (x_n), each element is a multiple of some element a_n and another b_n.

$$x_n = a_n b_n \tag{5.29}$$

For every $a \in A$ and $b \in B$, there exists a subsequence of (a_n) and (b_n) that converges to a and b respectively.

$$a_{n_k} \to a; b_{n_j} \to b \tag{5.30}$$

Since $a \in A$ and $b \in B$, $ab \in A \cdot B$. Due to the sequence limit laws, $a_{n_k} b_{n_j}$ converges to $a \cdot b$. This means that for any $a \cdot b \in A \cdot B$, there is a subsequence (x_{n_j}) of (x_n) that converges to $a \cdot b$. Therefore, $A \cdot B$ is compact.

5.6.3 Solution (c)

Suppose that $A = \left\{\frac{1}{2n}\middle| n \in \mathbb{N}\right\} \cup \{0\}$ and $B = \left\{2n+1\middle| n \in \mathbb{N}\right\}$. Both are closed. However, $A \cdot B$ contains a sequence that converges to 1 but does not contain it. This means that $A \cdot B$ does not contain all its limit points, so it is not closed.

5.7 Problem 5.22

Let A be the set of numbers in $[0, 1]$ whose decimal expansions use only the numbers 2, 5 and 8. For example, $0.5822585582 \in A$, and $0.2525252525\ldots \in A$. Prove that A is a closed set.

5.7.1 Solution

Proof by Contradiction. Suppose for contradiction that a sequence in A converges to a point $a \notin A$. Throughout, use the decimal expansion that does not end in an infinite string of 9's.

$$\forall\varepsilon > 0, \exists N : \forall n > N, |a_n - a| < \varepsilon \tag{5.31}$$

a and every a_n can be written in the form of a sum of numbers.

$$a_n = \sum_{k=1}^{n} s_k \times 10^{-k}, s_k \in \{2, 5, 8\} \tag{5.32}$$

$$a = \sum_{k=1}^{\infty} t_k \times 10^{-k}, t_k \in \{2, 5, 8, m\}, m \in \{1, 3, 4, 6, 7, 9, 0\} \tag{5.33}$$

We can put these into the convergence equation.

$$s_k \in \{2, 5, 8\}; t_k \in \{2, 5, 8, m\}; m \in \{1, 3, 4, 6, 7, 9, 0\} \tag{5.34}$$

$$\left| \sum_{k=1}^{n} s_k \times 10^{-k} - \sum_{k=1}^{\infty} t_k \times 10^{-k} \right| < \varepsilon \tag{5.35}$$

$$\left| \sum_{k=1}^{n} (s_k - t_k) \times 10^{-k} - \sum_{k=n+1}^{\infty} t_k \times 10^{-k} \right| < \varepsilon \tag{5.36}$$

Suppose that point j is the first point where $t_j \in m$ and $s_k \neq t_k$, while $s_j \in \{2, 5, 8\}$. We can simplify $\sum_{k=1}^{n}(s_k - t_k) \times 10^{-k}$.

$$\sum_{k=1}^{n} (s_k - t_k) \times 10^{-k} = \sum_{k=j}^{n} (s_k - t_k) \times 10^{-k} \tag{5.37}$$

$$\left| \sum_{k=j}^{n} (s_k - t_k) \times 10^{-k} - \sum_{k=n+1}^{\infty} t_k \times 10^{-k} \right| < \varepsilon \tag{5.38}$$

At point j, $|s_j - t_j| \geq 1$, which means that $s_j - t_j \geq 1$ or $t_j - s_j \geq 1$. This gives us two cases.

Case 1: $s_j - t_j \geq 1$. Take the above convergence equation. If $n > j$, then the maximum value of $s_k - t_k$ is 8. Even if for all $k \geq n+1$, $t_k = 9$, it will always be the case that the magnitude of the sum up to and including $k = n$ will be greater than the sum after that point.

$$\sum_{k=j}^{n}(s_k - t_k) \times 10^{-k} > \sum_{k=n+1}^{\infty} t_k \times 10^{-k} \tag{5.39}$$

$$\left|\sum_{k=j}^{n}(s_k - t_k) \times 10^{-k} - \sum_{k=n+1}^{\infty} t_k \times 10^{-k}\right| > 0 \tag{5.40}$$

$$\sum_{k=j}^{n}(s_k - t_k) \times 10^{-k} - \sum_{k=n+1}^{\infty} t_k \times 10^{-k} < \varepsilon \tag{5.41}$$

Our sum of sums will always be greater than or equal to 10^{-j}.

$$10^{-j} \leq \sum_{k=j}^{n}(s_k - t_k) \times 10^{-k} - \sum_{k=n+1}^{\infty} t_k \times 10^{-k} < \varepsilon \tag{5.42}$$

This creates a contradiction, since ε should be able to be less than 10^{-j}.

There is one edge case, when $j = n$, $s_j - t_j = 1$, and $t_k = 9$ for all $k > n$. In that case, $\sum_{k=n+1}^{\infty} t_k \times 10^{-k} = 0.\overline{99} \times 10^{-j} = 10^{-j}$. Therefore, as long as the decimal expansion terminates at digit j, we can increase t_j by 1. This would mean that $s_j = t_j$, which would in turn hold true for all k. This, however, would have $a \in A$, which would contradict our supposition for contradiction.

Case 2: $t_j - s_j \geq 1$. This will be pretty similar in execution to case 1. Take the above convergence equation. If $n > j$, then the maximum value of $t_k - s_k$ is 8. Even if for all $k \geq n+1$, $t_k = 9$, it will always be the case that the magnitude of the sum up to and including $k = n$ will be greater than the sum after that point.

$$\left|\sum_{k=j}^{n}(s_k - t_k)\times 10^{-k} - \sum_{k=n+1}^{\infty} t_k \times 10^{-k}\right| = \left|\sum_{k=n+1}^{\infty} t_k \times 10^{-k} - \sum_{k=j}^{n}(s_k - t_k)\times 10^{-k}\right| \tag{5.43}$$

$$\left|\sum_{k=j}^{n}(s_k - t_k)\times 10^{-k} - \sum_{k=n+1}^{\infty} t_k \times 10^{-k}\right| = \left|\sum_{k=n+1}^{\infty} t_k \times 10^{-k} + \sum_{k=j}^{n}(t_k - s_k)\times 10^{-k}\right| \tag{5.44}$$

$$\left|\sum_{k=n+1}^{\infty} t_k \times 10^{-k} + \sum_{k=j}^{n}(t_k - s_k) \times 10^{-k}\right| > 0 \tag{5.45}$$

$$\sum_{k=n+1}^{\infty} t_k \times 10^{-k} + \sum_{k=j}^{n}(t_k - s_k) \times 10^{-k} < \varepsilon \tag{5.46}$$

Our sum of sums will always be greater than or equal to 10^{-j}.

$$10^{-j} \leq \sum_{k=n+1}^{\infty} t_k \times 10^{-k} + \sum_{k=j}^{n}(t_k - s_k) \times 10^{-k} < \varepsilon \tag{5.47}$$

This creates a contradiction, since ε should be able to be less than 10^{-j}. There is no edge case this time.

Ergo, our initial assumption is false, so A is closed. OE∆

Scratch Work

A would be closed, also bounded so it'd be compact.

STE an open cover of A.

$$A \subseteq \bigcup_{\alpha=0}^{n} U_\alpha \tag{5.48}$$

$$\forall a \in A, \exists \delta > 0, \exists \alpha : (a - \delta, a + \delta) \subseteq U_\alpha \tag{5.49}$$

The numbers can be mapped by the natural numbers.

1	0.2
2	0.5
3	0.8
4	0.22
5	0.25
6	0.28
7	0.32
8	0.35
9	0.38

5.8 Problem 5.24

A set A is said to have the intersecting closedness property if it satisfies this condition: If S is any collection of closed sets for which the intersection of any finite number of sets from S contains an element of A, then the intersection of every set in S also contains an element of A.

Prove that A has the intersecting closedness property if and only if A is compact.

5.9 Problem 5.26

(a) Prove that if A is compact, then $\sup(A)$ exists and $\sup(A) \in A$. Does the same hold for the infimum?

(b) Give an example of a set which contains its supremum and its infimum, but is not compact.

(c) If a set contains its supremum, its infimum, and all of its limit points, must it be compact?

5.10 Problem 5.28

For which compact sets A does there exist an $m \in N$ such that every open cover of A contains a subcover containing at most m open sets?

5.11 Problem 5.30

Give two examples of sets which are neither open nor closed. Have one example be a bounded set and the other be an unbounded set.

5.12 Problem 5.31

(a) Find the interior, exterior and boundary for each of the following sets: $(0, 1)$, $[0, 1]$, $\mathbb{R}$ and $\mathbb{Q}$.

(b) Prove that for any set A, we have $\mathbb{R} = \mathrm{Int}(A) \cup \partial A \cup \mathrm{Ext}(A)$, and this is a disjoint union.

5.13 Problem 5.32

Prove that the only sets with empty boundary are $\mathbb{R}$ and $\emptyset$.

5.14 Problem 5.33

(a) Explain intuitively what it means for a set A to be connected.

(b) Give an example of a connected set and of a set that is not connected. You do not need to prove your answers.

(c) Give an example of a set A which is not connected, but $A \cup \{4\}$ is connected.

(d) Prove that $\{1, 2, 3, 4, 5\}$ is not connected.

(e) Prove that $\mathbb{Q}$ is not connected.

(f) Give an example of a set A which is not connected, but there exists some $x_0 \in \mathbb{R}$ such that $A \cup x_0$ is connected.

(g) Prove that a set of real numbers with more than one element is connected if and only if it is an interval.